Reproducibility in Data Science

Group 5

Introduction

Paper summary

The paper selected for this project, “Wine Feature Importance and Quality Prediction: A Comparative Study of Machine Learning Algorithms with Unbalanced Data,” focuses on using machine learning techniques to predict wine quality based on its physicochemical properties. The authors compare five models, Support Vector Machine (SVM), Random Forest (RF), Gradient Boosting (GB), K Nearest Neighbors (KNN), and Decision Tree (DT), to determine which performs best, while also examining which features have the strongest impact on wine quality. The motivation behind the study is that traditional wine evaluation relies on human experts, which can be subjective and time consuming. Machine learning provides a more scalable and consistent alternative. One of the key findings of the paper is that alcohol content is the most influential feature, and that SVM achieves the highest performance, reaching an accuracy of 96% when the dataset is balanced and the model is tuned.

Reproduction goals

The goal of this reproduction project was to replicate the main findings of the paper as closely as possible. Specifically, the objective was to rebuild the preprocessing pipeline, implement the same five machine learning models, and compare performance across imbalanced and balanced datasets. Another important goal was to evaluate whether SVM would still emerge as the best-performing model and whether the feature importance results, particularly the dominance of alcohol, could be reproduced.

Methodology

Reproduction approach

To carry out the reproduction, the workflow followed the general structure described in the paper. The red wine dataset from the UCI repository was used, which contains 1,599 observations and 11 input features related to chemical properties. The preprocessing steps included removing duplicates, handling missing values, and applying label encoding to convert wine quality into three categories: bad, normal, and

good. Feature selection was also performed, and the “residual sugar” variable was excluded since the original paper found it to have minimal impact on prediction performance. The data was then standardized to ensure that all features were on a similar scale before training the models. As the paper also states for the hyper parameter tuning, to address the imbalanced dataset, using SMOTE(Synthetic Minority Oversampling Technique) where using K nearest neighbors to get more samples that are synthetic data points to balance the dataset.

Environmental setup

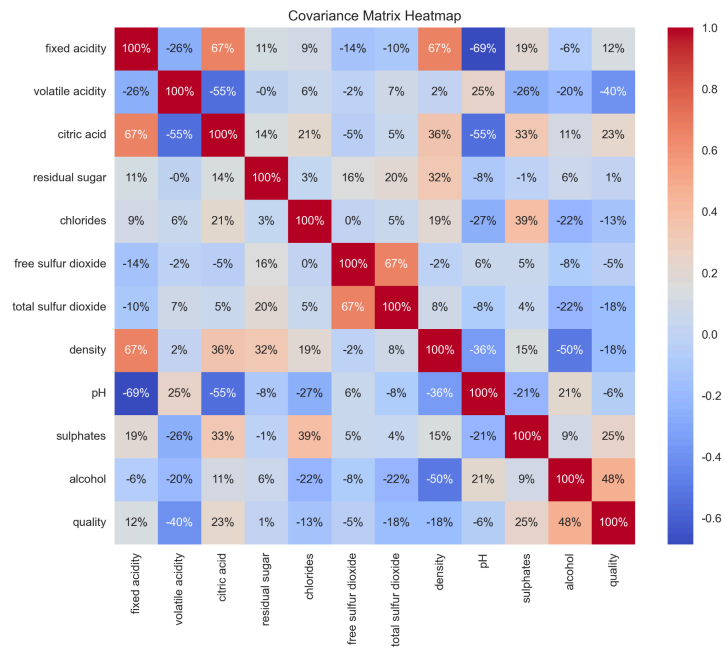
The project was completed using Python in a Jupyter Notebook environment. Common data science libraries such as pandas, numpy, scikit learn, matplotlib, and seaborn were used throughout the analysis. One challenge encountered during this step was that the original paper did not specify exact software versions, which may have introduced slight variations in results. The paper also did not state how they made the test train split for the modeling and the ratio they used for the models. For the models the paper did not specify any parameters for the base model or the parameters for the grid search.

Data processing steps

The dataset used contains just under 1,600 observations and several features describing the chemical composition of red wine. The data was first cleaned by removing duplicates and checking for missing values. The target variable was then grouped into three categories so the problem could be treated as a classification task. Feature selection was applied, and one variable was removed due to its minimal impact on performance. Finally, the data was standardized so that all features were on a similar scale before model training.

Results

Comparison with original

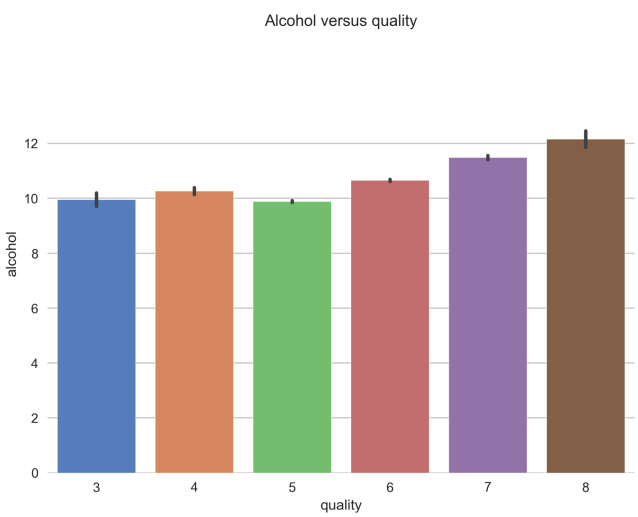


When attempting to reproduce the results presented in the original paper, the initial focus was on achieving consistency in the dataset, including its shape, features, and sparsity. The same preprocessing steps described in the study were followed, and the resulting summary statistics and visualizations such as bar charts and correlation heatmaps were reproduced almost exactly. This indicates that the exploratory data analysis phase was successfully aligned with the original work. Challenges arose during the modeling phase. While the reproduced results followed similar overall trends particularly the improvement in performance after addressing class imbalance the exact accuracy reported in the original paper was not consistently achieved. Performance generally remained in the mid-70% range, even after applying techniques such as SMOTE and conducting hyperparameter tuning through grid search. One of the primary difficulties in reproducing the results was the lack of specificity in the original methodology. There was noticeable ambiguity surrounding key components such as data scaling, train-test splitting, model hyperparameters, training procedures, cross-validation strategies, and evaluation methods. This lack of detail made it difficult to replicate the exact experimental setup. To address these uncertainties, additional steps were taken beyond the original description, including experimenting with different data splits, tuning hyperparameters, and testing alternative cross-validation approaches. While these efforts led to slight improvements, they did not fully bridge the gap between the reproduced results and those reported in the paper. In terms of model performance, both Support Vector Machine (SVM) and Random Forest emerged as strong performers across experiments. While SVM was identified as the top model in the original study, Random Forest often performed equally well or better in this reproduction. Despite this difference, the overall ranking of models and feature importance trends remained broadly consistent with the original findings.

Overall, while the general patterns and insights of the original study were successfully reproduced, exact performance replication was not achieved. These findings highlight the importance of detailed methodological transparency in machine learning research. Future work could involve conducting a more comprehensive grid search or further investigating data splitting and evaluation strategies to improve reproducibility.

Discrepancy analysis

The differences between the reproduced results and the original findings can likely be explained by missing details in the paper. Not all hyperparameters were clearly specified, and some preprocessing steps required assumptions. In addition, techniques used to balance the data involve randomness, which can lead to variation in results.



Differences in train test splits, random seeds, and software environments may have also contributed to these discrepancies. There is also a discrepancy even from the original paper where even with the same data class 0(bad wine) had high metrics for all the models even with a very small sample size. With only 63 for the red wine dataset it just seems mathematically implausible to get that high of metrics especially 1 for precision in the base model with recall having 0.2.

Ethics Discussion

Reproducibility challenges

This project highlighted several common challenges in reproducing data science research, many of which are discussed in the course readings. One of the biggest issues encountered was the lack of detailed information in the original paper. Important components such as exact hyperparameters, preprocessing decisions, and implementation details were either missing or not clearly explained. This made it difficult to fully replicate the results without making assumptions. This aligns with the ideas discussed in Leakage and the Reproducibility Crisis in ML-based Science, which explains how even small gaps in methodology or hidden assumptions can lead to results that are difficult or impossible to reproduce. In this project, even though the

general trends were replicated, the exact results varied, showing how sensitive machine learning workflows can be to these kinds of missing details.

Another challenge was related to randomness in the modeling process. Techniques such as sampling and data splitting can produce slightly different results each time they are run, especially if random seeds are not clearly defined. This connects to the reading Reproducible Data Science and Why it Matters, which emphasizes that reproducibility is not just about running code, but about being able to consistently achieve the same results under the same conditions. Without clear documentation of randomness and experimental setup, reproducibility becomes much more difficult to achieve.

Resource considerations

From a resource perspective, this project was manageable because the dataset was relatively small and did not require extensive computational power. However, the readings highlight that access to resources can be a major barrier in reproducibility. In Reproducible Data Science and Why it Matters, it is noted that differences in computing environments, software versions, and available tools can all impact results. Even though this project did not require advanced hardware, it still demonstrated how small differences in environment setup could influence outcomes. In larger or more complex studies, these issues would likely be even more significant.

Scientific integrity implications

Reproducibility is closely tied to scientific integrity, and this project reinforces why it is so important. If results cannot be reproduced, it raises concerns about their reliability and whether they can be trusted or applied in real-world settings. The reading Five Selfish Reasons to Work Reproducibly highlights that reproducibility is not just about benefiting others, but also about protecting the researcher’s own work and credibility. In this case, if the results of the wine quality prediction model could not be reasonably replicated, it would be difficult to confidently use those findings in practice.

Additionally, the FAIR Guiding Principles emphasize that data and research outputs should be findable, accessible, interoperable, and reusable. While the dataset used in this project was publicly available, the lack of shared code and full methodological transparency in the paper limited its reusability. This creates a gap between publishing results and actually enabling others to build on that work.

Overall, this project demonstrates that reproducibility is not just a technical challenge, but also an ethical responsibility. Clear documentation, transparency, and accessibility

are essential for ensuring that research can be trusted, validated, and extended by others.

Recommendations

Best practices

To improve reproducibility, researchers should clearly document all steps in their workflow, including preprocessing, model settings, and evaluation methods. Sharing code and data is also important, as it allows others to verify results directly. Setting random seeds and documenting the computational environment can help reduce variability in outcomes.

Process improvements

In addition to best practices, improvements can be made to the overall research process. Using version control systems helps track changes and maintain transparency. Writing clear and well organized code makes it easier for others to understand and reproduce the work. It is also important to document unsuccessful attempts, as they provide insight into the challenges encountered during the reproduction process.